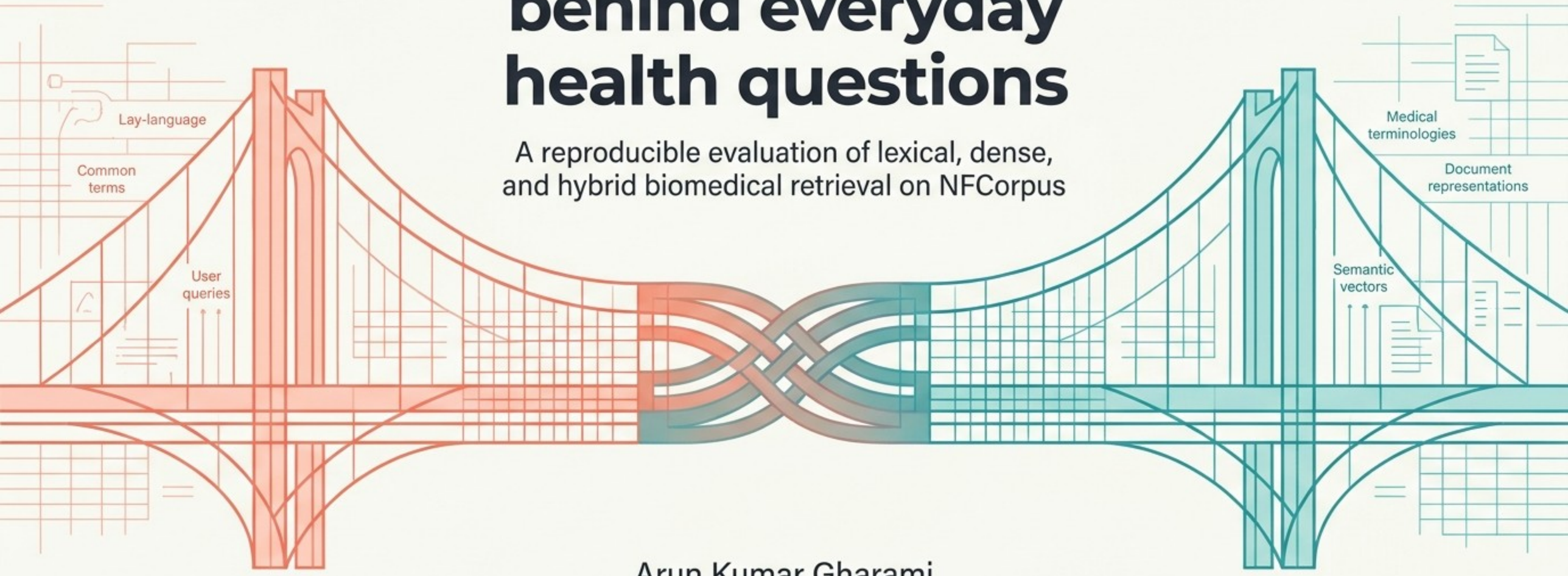
Finding the science behind everyday health questions

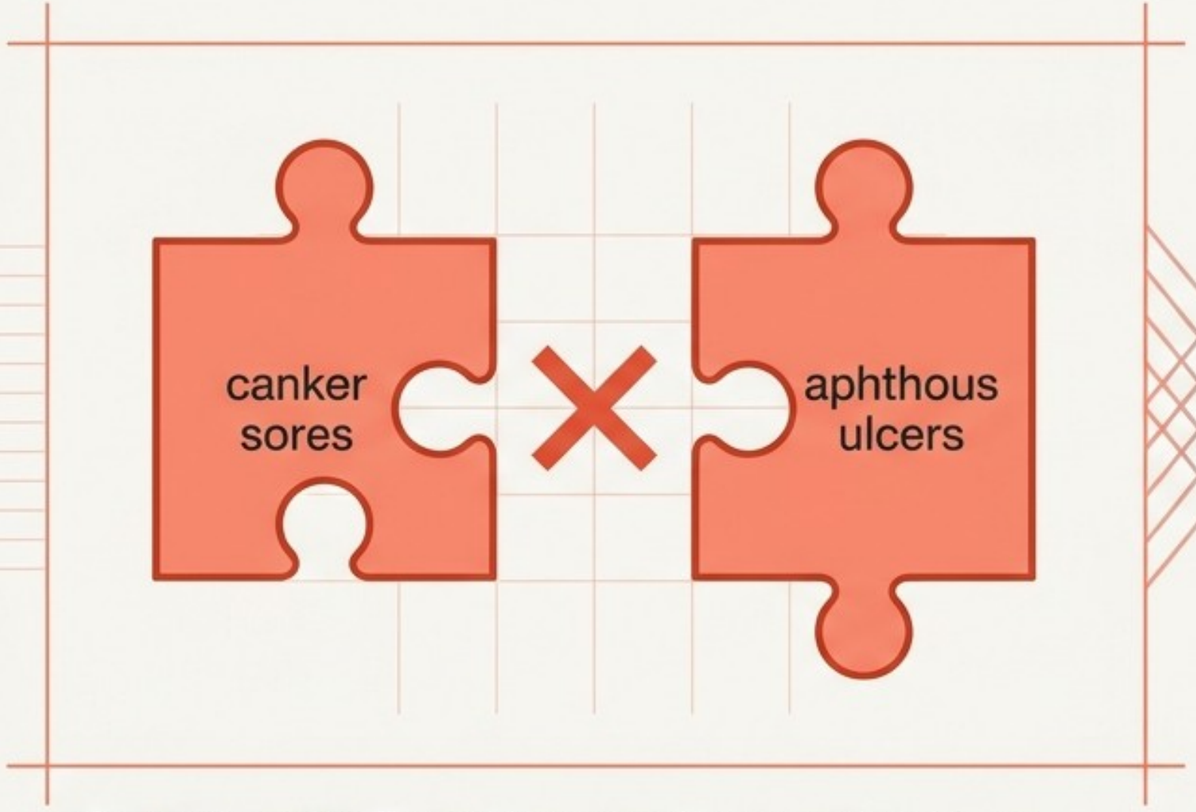
A reproducible evaluation of lexical, dense, and hybrid biomedical retrieval on NFCorpus



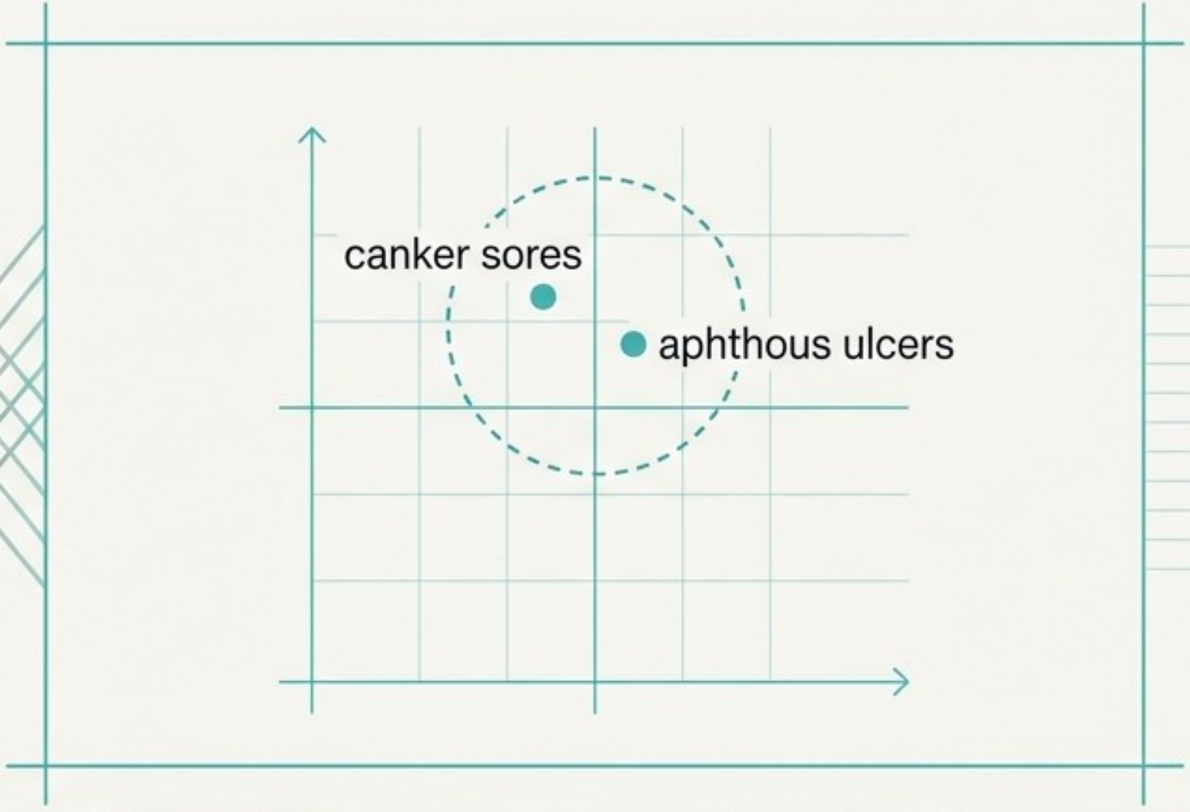
Arun Kumar Gharami

Gemini Notebook

Everyday users speak a different language than medical researchers



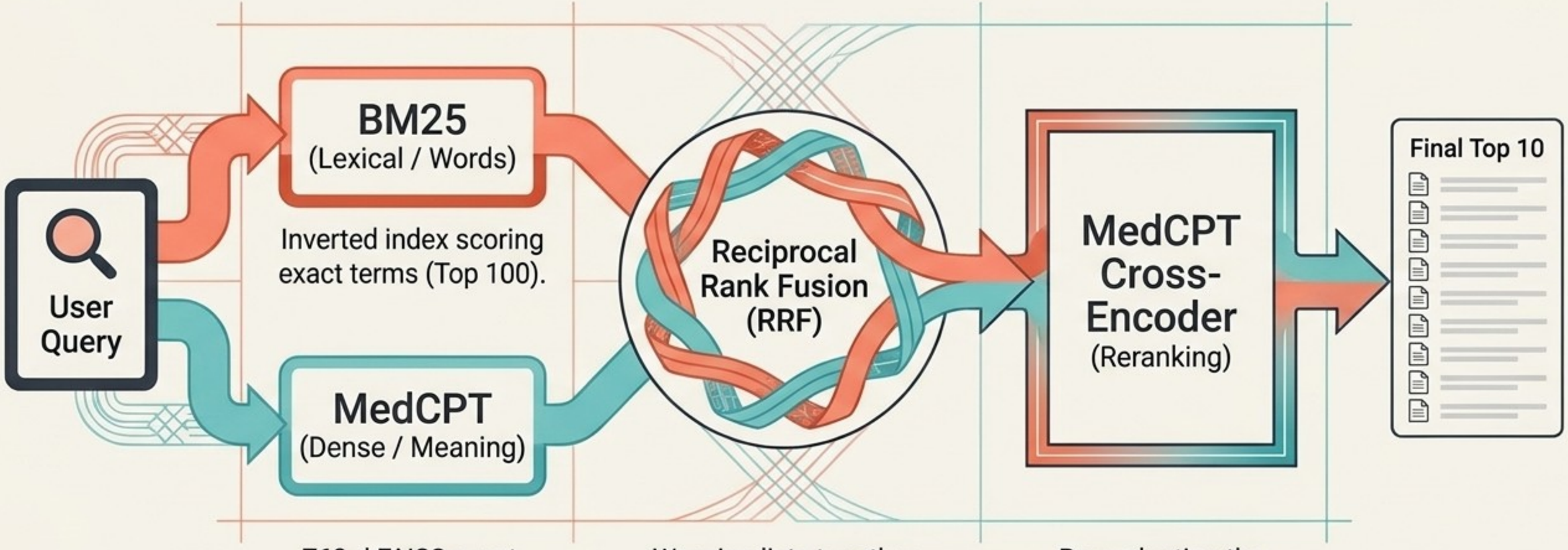
The Keyword Failure: The query 'Fosamax' yields an empty BM25 list because the relevant paper uses the generic term 'alendronate'. Zero shared words means zero lexical retrieval.



The Semantic Promise: Dense vector embeddings map meaning, not just characters, placing layperson terms and clinical terminology in the same mathematical neighborhood.

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Two retrievers, fused by rank, then re-read by a cross-encoder



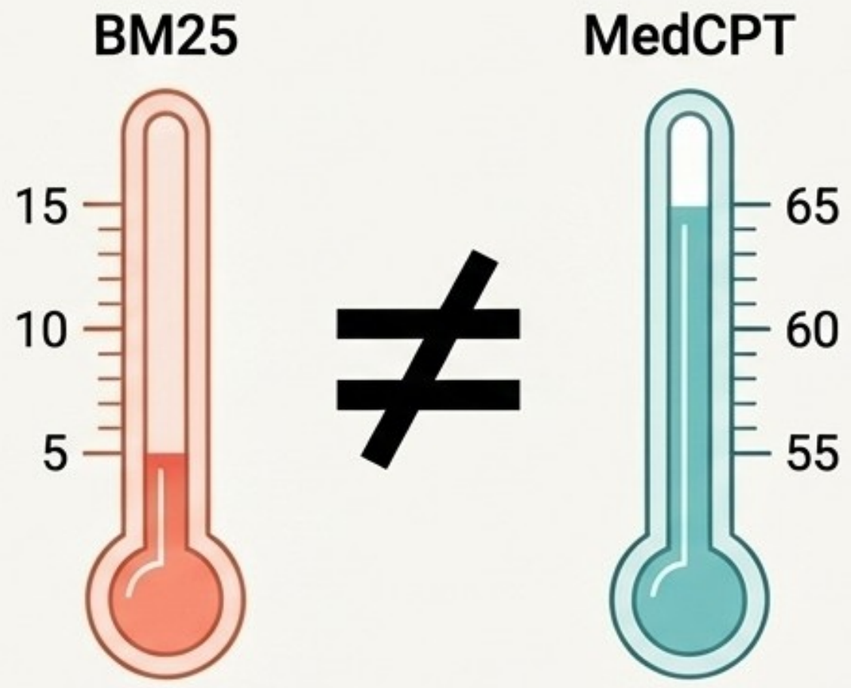
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System components balance precision, recall, and computational cost

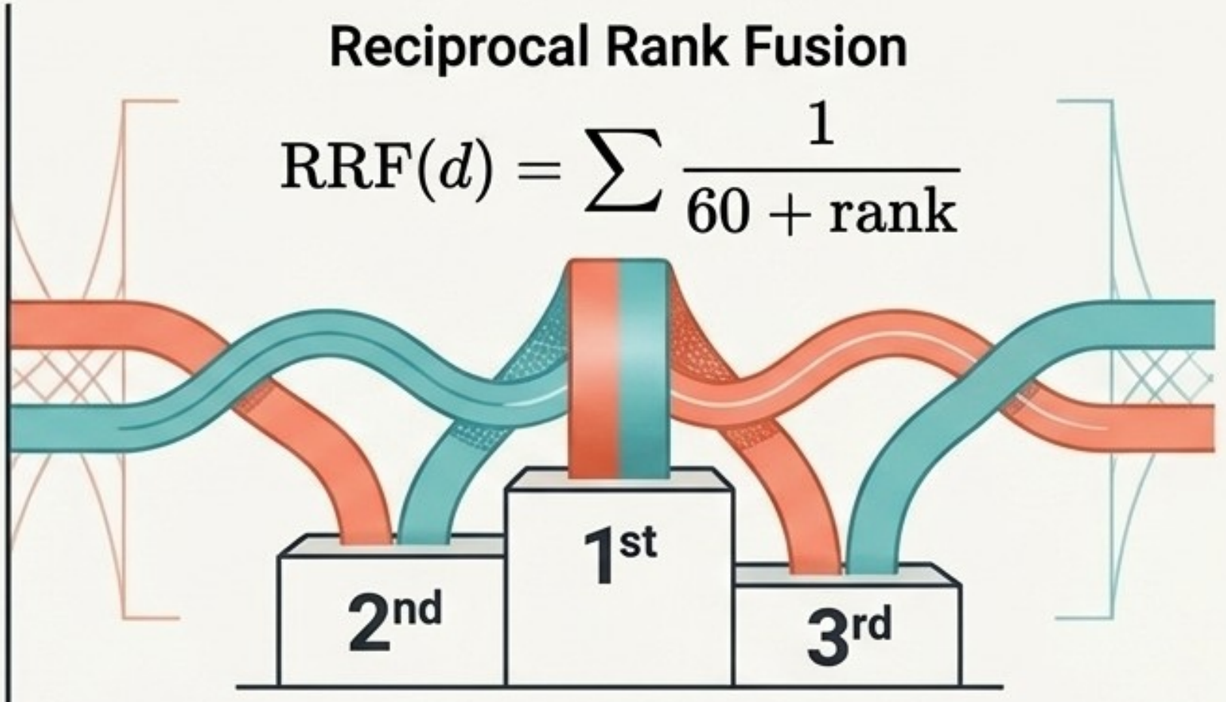
Retriever	Index Mechanism	Primary Strength	Critical Weakness	Latency (M1 Pro)
BM25 (Lexical)	Words (Inverted list)	Unbeatable exact-term precision	Vocabulary mismatch	2.2 ms
MedCPT (Dense)	Meaning (768-d Vector)	Semantic mapping across vocabularies	Semantic drift (loss of specific detail)	1.9 ms
Hybrid + Cross-Encoder	Combined + Attention	Holistic contextual understanding	Cannot surface documents outside the top 50	1,946 ms

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Fusing ranks instead of scores prevents one model from dominating



The Incompatibility: BM25 top scores are 5–15. MedCPT inner products are 55–65. Adding them directly would allow semantic search to silence lexical search.

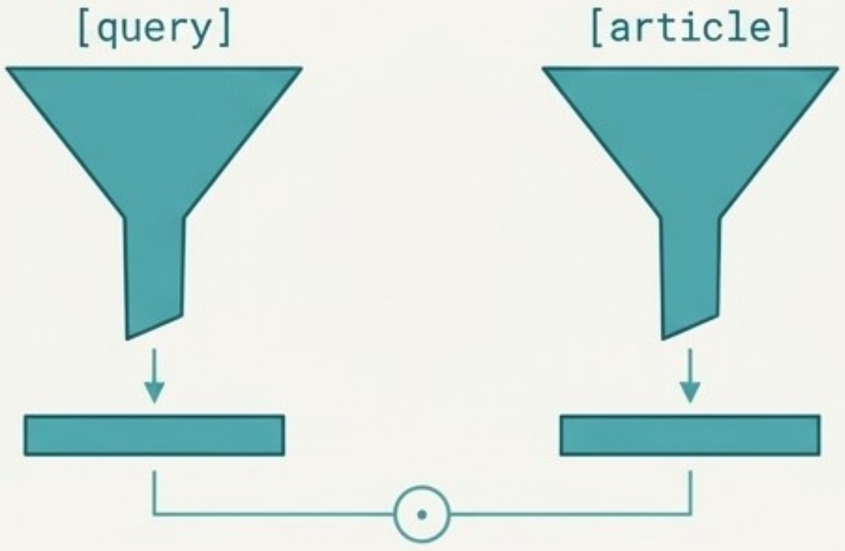


The Result: Agreement between both retrievers earns a higher final score than a single first-place finish in just one.

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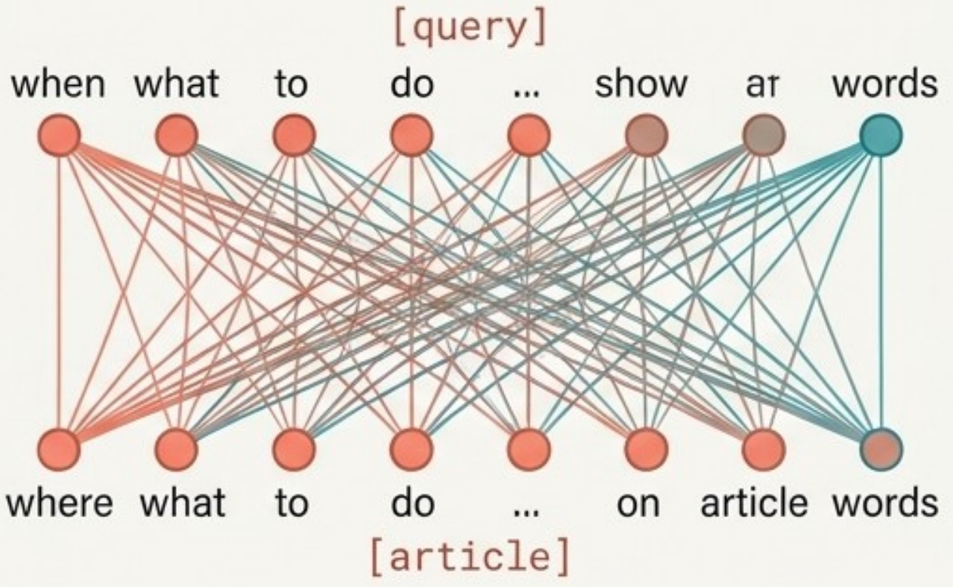
Cross-encoders read the query and document together at massive computational cost

Bi-Encoders (MedCPT Retrieval)



Compress queries and articles separately.
Highly efficient for searching millions of documents via FAISS.

Cross-Encoders (Reranking)



Feed both query and article through one transformer simultaneously.
Every query word attends to every article word.

The Trade-off: Reordering just the top 50 candidates takes ~480× longer than the entire retrieval phase.

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The Crucible: Evaluation on the NFCorpus Benchmark

3,633 PubMed titles/abstracts evaluated against 323 held-out lay-language test queries. Graded relevance (0, 1, 2).

P@10



How much of the first page is useful? (Regardless of position).

MRR@10



How quickly does the user hit the first useful result?

nDCG@10



Does the system rank highly relevant (grade-2) papers above marginally relevant (grade-1) papers?

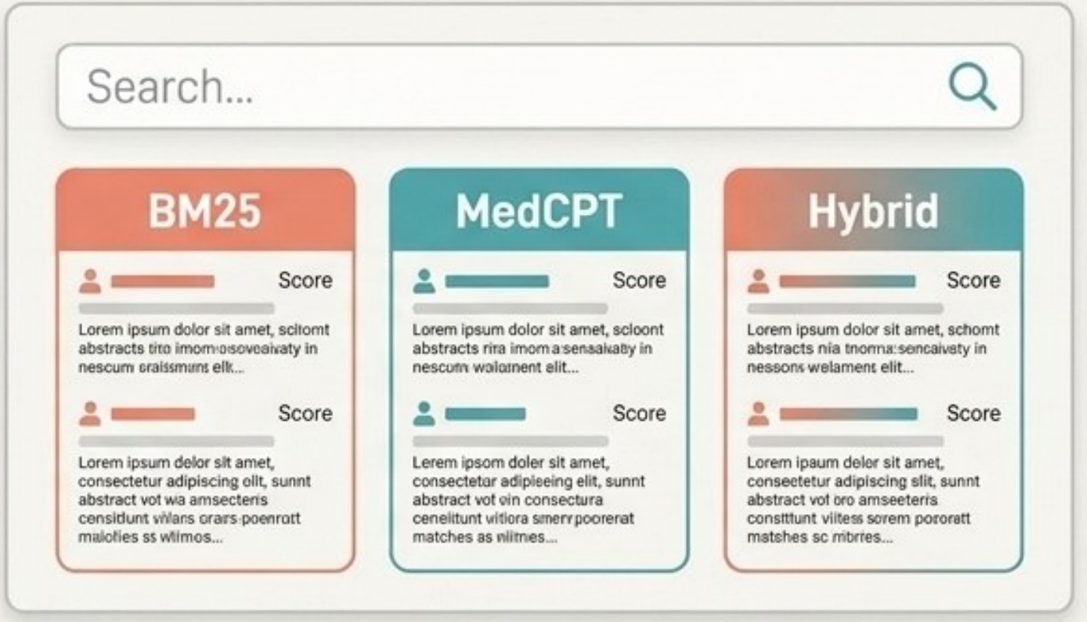
Methodology Constraint: Zero tuning on the test split. BM25 uses textbook defaults (k1=1.2, b=0.75). Paired bootstrap testing applied ($p < 0.05$).

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Built for absolute reproducibility and live demonstration

```
src/biomedical_ir/
├── scripts/
└── tests/
```

Robust Foundation: Engineered as a reusable Python library with 248 passing unit tests, including hand-computed BM25/RRF edge cases.



Live Research Portal: The same functions driving the evaluation scripts power a live interactive UI. The demo runs real queries, allowing qualitative comparison of exact keyword matches versus semantic drift in real-time.

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Dense retrieval is the major leap; complex reranking is a costly minor step

Model	P@10	Recall@100	MRR@10	nDCG@10	Latency (ms)
TF-IDF	0.217	0.237	0.506	0.305	3.6
BM25 (Defaults)	0.207	0.230	0.494	0.295	2.2
BGE (General Dense)	0.280	0.337	0.556	0.371	2.7
MedCPT (Biomedical Dense)	0.270	0.349	0.549	0.365	1.9
Hybrid (BM25+MedCPT RRF)	0.260	0.339	**0.568**	0.362	4.1
+ Cross-encoder (Pool 50)	**0.277**	*0.278*	0.567	**0.373**	1,946

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Interpreting the data: Surprises and Statistical Truths

TF-IDF beats BM25

A surprise finding (nDCG@10 $p = 0.03$). Untuned BM25 defaults ($b=0.75$) may over-penalize the short PubMed abstracts in this corpus.

The Dense Victory is Absolute

Both BGE and MedCPT crush BM25 on all five metrics ($p < 0.005$). Vocabulary mismatch is the primary failure mode of this dataset.

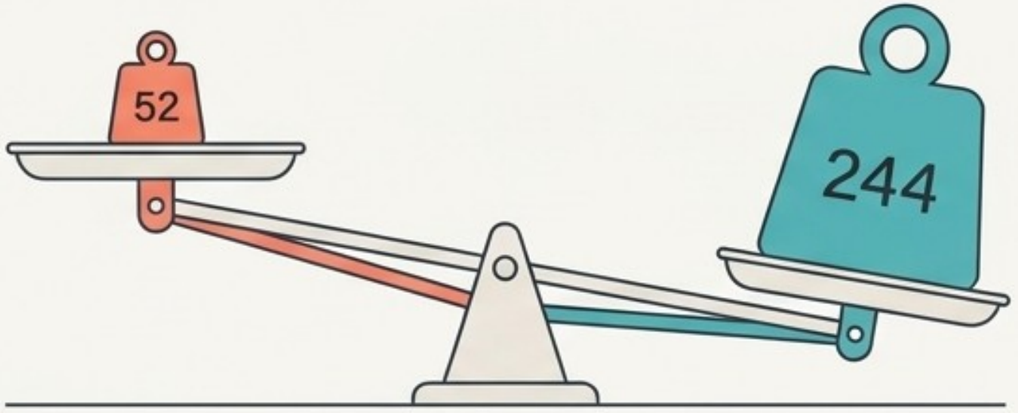
Reranking's Diminishing Returns

The nDCG@10 gain from the Cross-Encoder is the highest point estimate, but it is not statistically significant ($p = 0.19$) despite costing 480× the latency.

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Diagnosing failures across 323 test queries

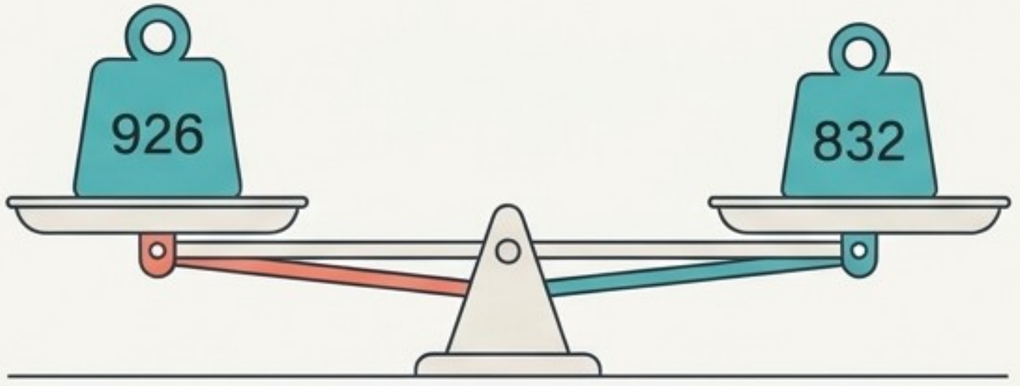
BM25 rescues relevant docs that MedCPT missed.



MedCPT rescues relevant docs that BM25 missed.

Lexical Failures Dominate: The vocabulary gap is heavily asymmetric.

Reranker moves relevant docs UP



Reranker moves relevant docs DOWN

The Reranker Coin Flip: The cross-encoder is highly volatile. It actively demotes almost as many good documents as it promotes.

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Qualitative reality: Exact precision vs. Semantic drift

The Lexical Win

Query: salmon



BM25: Success.

Finds exact salmon paper at Rank 2.

MedCPT: Failure.

Suffers “semantic drift”, returning generic seafood/fish papers, missing the exact subject entirely.

The Semantic Win

Query: canker sores



BM25: Failure.

Returns a completely blank list.

MedCPT: Success.

Instantly bridges the lay-language query to the clinical term “aphthous ulcers” at Rank 1.

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“Meaning-based retrieval fixes the vocabulary gap, but complex fusion and reranking must earn their their computational cost.”

- Dense retrieval is mandatory for lay-to-clinical information systems.
- Reciprocal Rank Fusion creates a highly capable, balanced system with near-zero latency penalty.
- Cross-encoder reranking introduces massive computational overhead that, on this corpus, does not yield statistically significant gains.
- Next Steps: Dev-set parameter tuning and validation on multi-benchmark datasets (SciFact, TREC-COVID).

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